

Server Full-Suite Regression Evidence — 2026-07-10

Revision: 1 **Last modified:** 2026-07-09T20:14:32Z **Scope:** server/ Go buildable project (Stream FS, §11.4.119 exclusive owner) **Authority:** §11.4.40 (full-suite retest slice), §11.4.4 (no-regression), §11.4.169 (comprehensive test types), §11.4.50 (determinism), §11.4/§11.4.6/ §11.4.69 (anti-bluff, cited evidence)

Purpose

Fresh comprehensive green regression evidence for the server on the current HEAD, confirming this session’s server changes integrate cleanly with the whole suite:

- `internal/api/memory_test.go` — memory-leak discriminator rewrite (review remediation)
- `internal/api/token_bench_test.go` — new token benchmarks

Environment

Field	Value
Repo	github.com/HelixDevelopment/helix_ota/server
Branch	main
HEAD SHA	1df9a649d627e21109f4bd6b1b8a8646cb4b2f40
HEAD subject	test(server): fix memory-leak discriminator (review remediation) + security/benchmark evidence
Go toolchain	go1.26.4-X:nodwarf5 linux/amd64
CPU	AMD Ryzen Threadripper 7970X 32-Cores
Run timestamp (UTC)	2026-07-09T20:14:32Z

Target files confirmed present + committed at HEAD (clean working tree for both): internal/api/memory_test.go (15688 B), internal/api/token_bench_test.go (3202 B).

Per-step results

#	Step	Command	Result	Evidence log
1	HEAD	git rev-parse HEAD	1df9a649... (see above)	—
2	Build	go build ./...	PASS — exit 0, no diagnostics (0-byte log = clean)	logs/02_go_...
3	Vet	go vet ./...	PASS — exit 0, no diagnostics (0-byte log = clean)	logs/03_go_...
4	Full suite	go test ./... -count=1	PASS — exit 0, every package ok (2 pkgs [no test files])	logs/04_go_...
5	Race	go test -race -count=1 ./internal/api/... ./internal/store/... ./internal/rollout/...	PASS — exit 0, grep -c "DATA RACE" = 0	logs/05_go_...
6a	Memory tests	go test -run 'TestMemoryGrowthClassifier\ TestMemory_SustainedAPILoadNoGrowth' -count=1 -v ./internal/api/	PASS — exit 0	logs/06a_me...
6b	Token benchmarks	go test -bench=Token -benchmem -run='^\$' -count=1 ./internal/api/	PASS — exit 0	logs/06b_to...
7	Determinism (-count=3)	go test -run '...' -count=3 ./internal/api/	PASS — exit 0, 1 ok, 0 FAIL	logs/07_det...

#	Step	Command	Result	Evidence lo
7b	Determinism (3 invocations)	3× go test -run '...' -count=1 -v ./ internal/api/	PASS — 3×3 exit 0, identical PASS census, 0 FAIL	logs/ 07b_determi

Step 4 — full non-integration suite (all ok)

```

ok  cmd/applyport 0.924s      ok  cmd/ota-device-emu 0.924s
ok  cmd/ota-server 1.690s
ok  internal/api 6.957s      ?   internal/api/manager-dist [no
test files]
ok  internal/config 0.002s    ok  internal/device 0.030s
ok  internal/deviceemu 0.181s
ok  internal/fabric 0.004s    ok  internal/health 0.002s
ok  internal/rollout 0.003s
ok  internal/store 0.004s     ok  internal/transport 0.178s
ok  tests/chaos 0.026s       ok  tests/stress 0.012s
ok  tools/loadtest 1.365s

```

Step 5 — race detector (0 data races)

```

ok  internal/api 12.952s      ?   internal/api/manager-dist [no test
files]
ok  internal/store 1.080s     ok  internal/rollout 1.012s

grep -c "DATA RACE" → 0.

```

**Memory-test real numbers (§11.4.169
memory + §11.4.107(10) self-validated
analyzer)**

Test	Result	Real numbers
TestMemory_SustainedAPILoadNoGrowth	PASS (0.11s)	retainedSmall=4240 leak=false — reason noiseFloor=524288 (retention did not s

Test	Result	Real numbers
TestMemoryGrowthClassifier	PASS (0.00s)	Evidence: qa-result memory_sustained_ap table-driven classifie
TestMemoryGrowthClassifier_DetectsInjectedSteadyLeak (golden-bad discriminator)	PASS (0.06s)	injected steady leak retainedSmall=81536 leak=true (retainedI scaled with request caught)

The discriminator proves the rewritten classifier is **not** a PASS-bluff: it flags an injected 4 KB/iter leak as leak=true (golden-bad FAILs the leak-check as it must), while the real sustained-load workload is leak=false (golden-good PASSes).

Token benchmark numbers (internal/api, AMD Ryzen Threadripper 7970X)

Benchmark	iters	ns/op	B/op	allocs/op
BenchmarkTokenMint-2	1,362,794	803.5	1312	14
BenchmarkTokenVerify-2	661,659	1682	1240	22
BenchmarkTokenVerifyReject-2	2,445,129	481.4	768	10

All benchmarks completed PASS / ok.

Determinism (§11.4.50)

- -count=3 single run: exit 0, 1 ok line, 0 FAIL.
- 3 separate -count=1 -v invocations: all exit 0; PASS census across the 3 runs — TestMemory_SustainedAPILoadNoGrowth PASS ×3, TestMemoryGrowthClassifier(+variant) PASS ×6, FAIL count = 0, ok lines = 3. Identical outcome every run.

Out-of-scope (explicit SKIP)

- **Integration suite (-tags integration)** — NOT run. It requires podman + a live Postgres (files: internal/store/postgres_integration_test.go, internal/rollout/

- postgres_integration_test.go, ...fault..., ...coverage...,
faultproxy_test.go, pg_itest_lock_test.go, etc.). Owned by a
separate concern; §11.4.3 topology SKIP-with-reason, not a PASS-by-
default. This slice covers the default (non-integration) build only.
- No git write operations performed (conductor commits). Only read-only git used.

Verdict

GREEN. Build + vet clean (exit 0, no diagnostics); full non-integration suite all ok (exit 0); race detector 0 data races across internal/api + internal/store + internal/rollout; the session's memory-leak discriminator and token benchmarks pass on current HEAD with real captured numbers; determinism holds across -count=3 and 3 separate invocations. The session's server changes integrate cleanly with the whole suite — no regressions found.